

Zakhar Pashkin

Senior ML Engineer | Computer Vision, Document AI & Agentic Systems

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Senior ML Engineer with 8+ years developing computer vision and applied ML from research through deployment. Shipped food-recognition and OCR workflows at Carb Manager and built financial-document recognition at CFT. Currently in Riverstart's R&D ML team, developing document AI and evaluating engineering-analysis methods. Also maintain an AI nutrition service and publish model-profiling and optimization tools.

EXPERIENCE

Senior ML Engineer

Riverstart | Jul 2026 - Present

R&D ML | Remote

- Own R&D implementation and evaluation of an on-premises document assistant that extracts case facts and reconciles records for specialist review. Separate deterministic checks from model explanations and link outputs to source documents.
- Build evaluation workflows for scan registration against CAD references, CAD projection and drawing comparison, using controlled geometry to check projected and hidden edges. Develop construction-document analysis that links extracted quantities to source pages.
- Build repeatable evaluation and release workflows for LLM services with versioned artifacts, regression checks, observability and rollback.

ML / Computer Vision Engineer - Independent Projects

Dermaself & Calorio (client projects) | Jun 2024 - Present (project activity)

Ongoing Calorio maintenance | Remote

- Developed skin-analysis CV pipelines for Dermaself, covering guided capture, facial regions, pore and wrinkle segmentation, model evaluation and mobile/API integration with PyTorch, OpenMMLab, ONNX and Flutter.
- Resolved model-asset and runtime differences between Cloud Run and GPU deployments, restoring matching wrinkle-segmentation outputs in regression comparisons and profiling remaining bottlenecks.

Senior Computer Vision Engineer

Wombat Apps LLC / Carb Manager | Jun 2022 - Jun 2024

Contract | Remote

- Built and shipped food-recognition and nutrition-label workflows across cloud, iOS and Android for Carb Manager, a nutrition product serving millions of customers.
- Owned data preparation, model development and evaluation through APIs, mobile export and runtime profiling. Combined detection, OCR and table parsing into structured nutrition data, working with mobile, backend and product teams on quality and failure handling.

EARLIER EXPERIENCE

Computer Vision Engineer

CFT | Jun 2019 - Jun 2022

Financial technology

- Built core detection, segmentation and OCR models for CFT's financial-document recognition platform using PyTorch and OpenMMLab.
- Optimized meter-recognition models for mobile use through architecture changes, quantization and runtime profiling. Built active-learning and assisted-annotation workflows to reduce manual labeling, and mentored engineers on data quality and evaluation.

Machine Learning Specialist & Project Mentor

School of AI | Oct 2018 - Oct 2019

Part-time community program

- Built ML learning software, coordinated remote contributors and mentored applied CV projects from dataset definition through evaluation and working demos.

SELECTED PROJECTS & RESEARCH

Agnitra | Published on PyPI

Published a Python SDK and CLI for model profiling and inference optimization, combining runtime telemetry, baseline comparisons and optimization passes, including hardware-aware quantization for decoder LLMs.

Calorio / YourFit | Maintained AI nutrition service

Built and maintain a Telegram AI nutrition service for meal logging from photos, voice and text, with a food diary, nutrition-goal tracking, deployment and technical support.

InQuest | Document-agent engineering

Built document agents using OpenAI Agents SDK and MCP, with retrieval, tool calls, handoffs and persistent context across document collections.

Multimodal Video Search | Retrieval R&D

Designed video retrieval combining visual embeddings, speech transcripts, on-screen text and scene descriptions with dense and sparse search to find relevant clips.

LigninQC | Research tooling | In progress

Built a reproducible literature-discovery and audit pipeline for computational lignin research, with publication-version grouping and field-level source provenance.

TECHNICAL EXPERTISE

Computer vision: Python, PyTorch, OpenMMLab (MMEEngine, MMCV, MMDetection, MMSegmentation, MMOCR), OpenCV, YOLO, OCR, segmentation, CLIP/VLMs, ONNX Runtime, Core ML

LLMs & agents: LangGraph, LangChain, OpenAI Agents SDK, MCP, Pydantic, LlamaIndex, RAG, Qwen/vLLM, Qdrant, Neo4j, pgvector, tool calling, evaluation and human review

ML systems: FastAPI, PostgreSQL, Redis, Docker, Kubernetes, AWS/GCP, Cloud Run, GitLab CI/CD, GitHub Actions, MLflow/ClearML, Prometheus/Grafana

EDUCATION

Siberian Federal University | Computer Science & Microelectronic Engineering | 2007-2012